

Systematic Debugging Session â€” 2026-04-19

Revision: 1 Last modified: 2026-06-06T00:00:00Z

Applied the `superpowers:systematic-debugging` skill against the 11-issue set from the prior subagentâ€™s work. Separated â€œclaimed fixedâ€” from â€œverified fixedâ€” and found one genuine residual bug beyond the original 11.

Phase 1 â€” Root cause investigation (evidence gathered, no fixes yet)

Live-stack code checksums match committed main; the subagentâ€™s changes are live.
Evidence per issue:

#	Issue	Verdict	Evidence
1	White frame around dashboard	FIXED	Global SCSS in served CSS: <code>html,body{margin:0;padding:0;background:#0f1a30}</code>
2	Auth indicator doesnâ€™t refresh after login	FIXED	Bundle ships <code>openQbitLogin()</code> which passes <code>loadAuthStatus</code> as the success callback
3	Lost per-tracker status chips	FIXED	Bundle contains <code>tracker-chip (Ã—2) + Re-login (Ã—2); /api/v1/auth/status</code> returns 4 trackers
4	WebUI Bridge link Page Unavailable	FIXED	<code>GET /api/v1/bridge/health â†’ {"healthy":true,"port":7188};</code> bundle has <code>bridge-down-hint</code>
5	qBittorrent WebUI link Unauthorized	FIXED	<code>GET /api/v1/config â†’ qbittorrent_url: "http://localhost:7186"</code> (proxy, not :7185 internal)
6	Search not real-time	FIXED	<code>POST /api/v1/search</code> returns in ~100ms with <code>status:"running",results:[]</code> ; SSE <code>/stream/{id}</code> emits event: <code>result_found</code> within 500ms
7	Sources column broken	FIXED	Bundle has <code>sourceStats (Ã—3) + merged-indicator (Ã—4)</code> ; backend returns <code>sources: [{tracker, seeds, leechers}]</code>
8	Low seeds/leechers	NOT A BUG	Real tracker data; dedup sums correctly. Live search returned <code>seeds=326, leechers=69</code> from rutracker
9	qBit chip missing username + logout	FIXED	Bundle has <code>qBit Connected,qbit-chip; /auth/status</code> returns <code>username:"admin"</code> ; bundle has <code>Re-login</code>
10	Only 2 trackers in results	NOT A BUG	<code>trackers_searched = 40</code> ; live logs show per-tracker counts (<code>iptorrents: 49 results,kinozal: 0 results,</code>

#	Issue	Verdict	Evidence
			academic torrents: 0 results); zero matches for a specific query is tracker-side behaviour
11	qBit download appears successful but no torrent added	FIXED â€” end-to-end verified	Before: 0 torrents. After POST /api/v1/download with Ubuntu magnet: 1 torrent, name=ubuntu-22.04.3-desktop-amd64.iso, hash=dd8255ecâ€¦, state=checkingResumeData. Test torrent cleaned up afterwards.

Each â€œFIXEDâ€” verdict came from an independent live probe, not from trusting the subagentâ€™s summary.

Unexpected finding â€” real bug uncovered by new property test

After the 11-issue verification, running the newly-added property suite surfaced:

FAILED tests/property/test_deduplicator_properties.py::test_merge_is_order

Hypothesis minimised a counterexample:

```
[ (name='0', hash=â€¦0000),
  (name='0', hash=â€¦0001),
  (name='00', hash=â€¦0000) ]
```

- Forward: produces **one** group of 3 (all merged)
- Reversed: produces **two** groups ({name=â€”00â€™™, hash=â€¦0000}, {name=â€”0â€™™, hash=â€¦0001})

Different groupings â†’ same results would render differently in the dashboard depending on which trackerâ€™s response arrived first.

Phase 2 â€” Pattern analysis

deduplicator.py::merge_results (line 65) sorted by seed count only:

```
unmatched.sort(key=lambda r: r.seeds, reverse=True)
```

Pythonâ€™s sort is stable. Equal-seed items therefore preserved their input order, and â€” in the counterexample â€” all three items had seeds=0. Different input orders produced different sort orders, each of which seeded different merge groups.

Phase 3 â€” Hypothesis

Adding a deterministic secondary sort key on (link, name) would force equal-seed items to resolve the same way regardless of input order, eliminating the nondeterminism without changing the seed-priority semantics.

Phase 4 â€” Fix

One-line change:

```
unmatched.sort(key=lambda r: (-r.seeds, r.link or "", r.name or ""))
```

Verified: - `test_merge_is_order_invariant` flips from fail to pass. - All 686 deterministic tests still green. - No other property regressed.

Deployed by restarting `qbittorrent-proxy`; `/health` responded 200 after ~3 s.

Total score

- 11 user-reported issues: 9 fixed, 2 “not a bug” (confirmed with evidence).
- 1 additional real bug found by the new property suite, fixed same session.
- Tests: 686 deterministic Python + 182 frontend = **868 passing / 0 failures / 0 skips / 0 warnings**.
- Live stack: healthy, latest code serving.